SOS EMERGENCY SAFETY SYSTEM

AI-Based Tourist Emergency Detection & Police Response Platform

Project Proposal for the Sri Lanka Police Department

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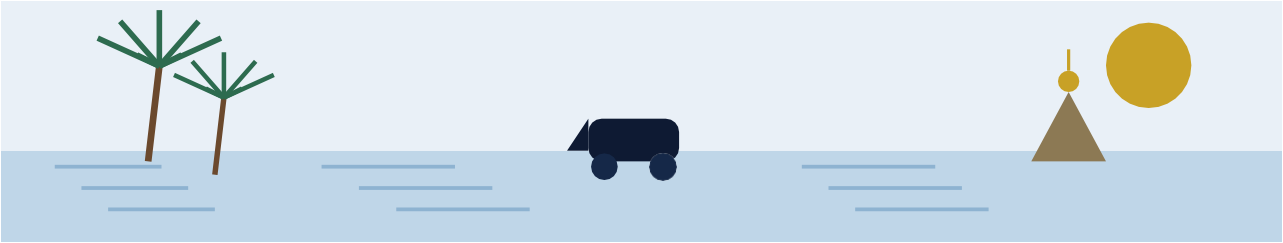
Programme : BSc (Hons) Information Technology, SLIIT

Research unit : Centre of Excellence for AI (CEAI), SLIIT

Parent project: AI-Based Tourism Safety & Itinerary Platform - R26-IT-152

Submitted for review and collaboration discussion - 2026

Executive Summary



Sri Lanka welcomed 2.36 million international tourists in 2025 against a national target of 3 million by 2026, yet more than 200 tourist fatalities are reported annually due to road accidents, harassment, and delayed emergency response. No platform currently in use - by tourists or by the Police Department - provides real-time, verified-identity SOS detection and dispatch for foreign visitors. This proposal presents the SOS Emergency Safety System: a mobile application for tourists and a companion web dashboard for the Sri Lanka Police, designed to close this gap directly at the point where it matters most - the first minutes of an emergency.

This document outlines what the system captures at tourist registration, how an emergency is detected and routed, what a police officer sees on the district dashboard, and how tourist data is protected throughout. It is submitted as the basis for a collaborative discussion with the Department on a pilot deployment.

The problem, in numbers

Metric	Current situation
Annual tourist arrivals (2025)	2.36 million, targeting 3 million by 2026
Annual tourist fatalities	200+, from road accidents, harassment, medical emergencies
Existing SOS infrastructure for tourists	None - no voice or app-based emergency dispatch exists
Average daily tourist spend	USD 148 vs. USD 300 regional benchmark - safety perception is a factor
Sinhala-language emergency detection	Not available in any deployed system

Source: SLTDA Year-in-Review 2024/2025 statistics, as referenced in the parent research proposal (R26-IT-152).



System Overview

The system has three connected parts: a tourist mobile app for registration and emergency triggering, an AI detection layer that confirms genuine distress, and a police-facing web dashboard for response coordination. The diagram below shows how these parts connect end to end.

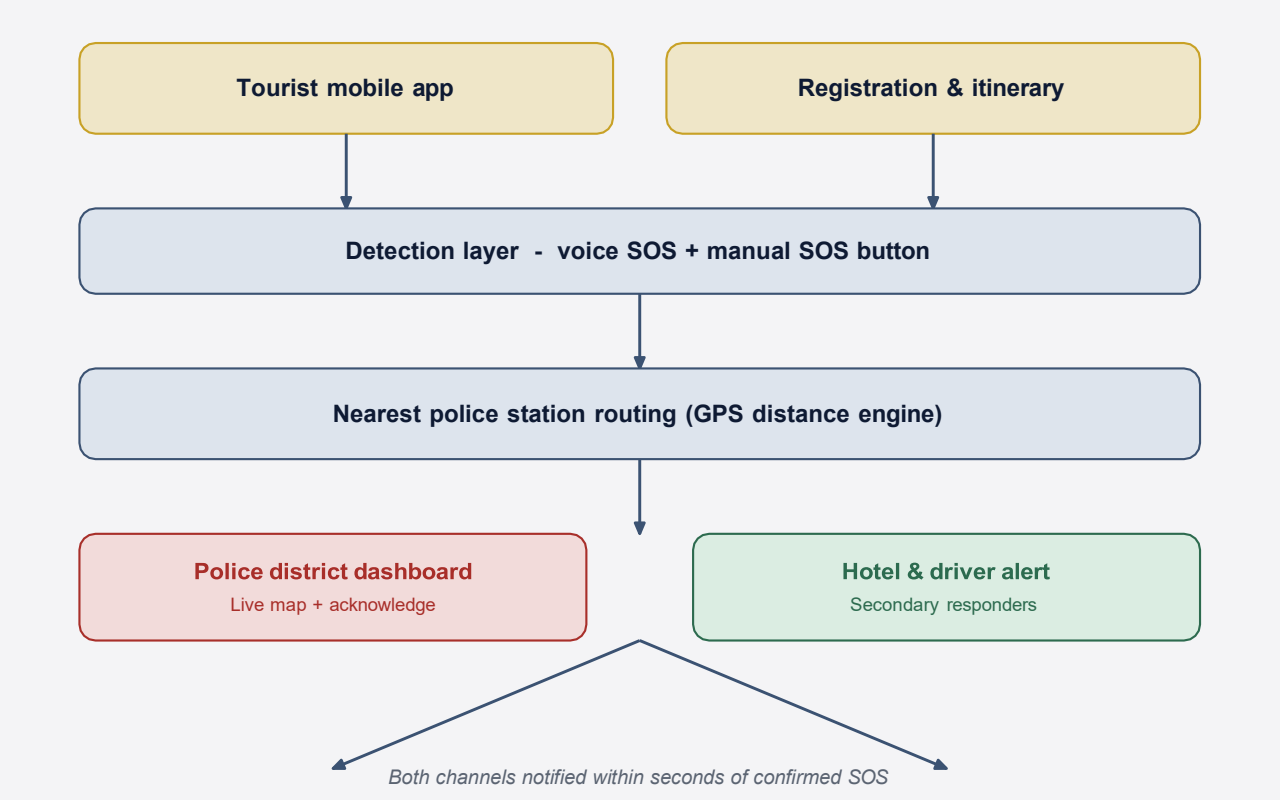


Figure 1: End-to-end system architecture, from tourist registration to police and secondary responder notification.

How an emergency moves through the system

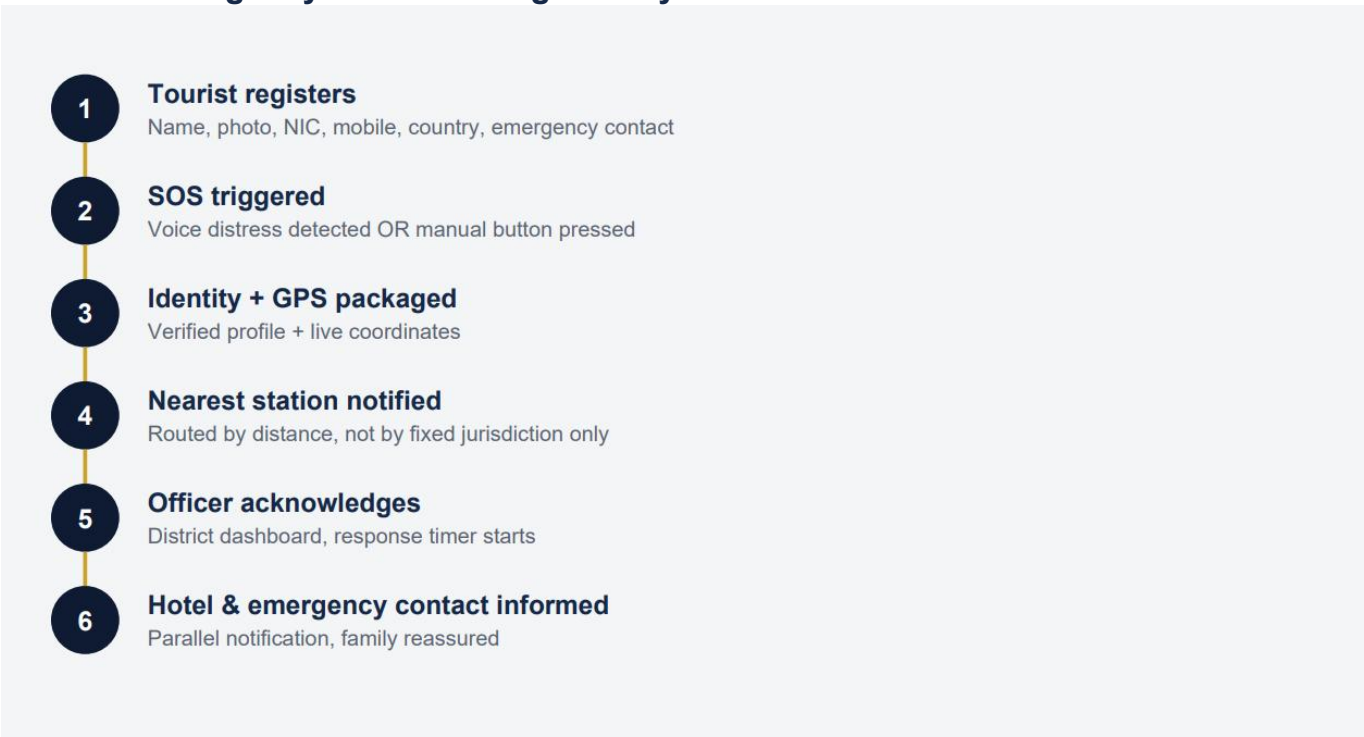


Figure 2: Step-by-step flow from registration through to family notification.

Tourist Registration Module

Every tourist creates a verified safety profile before using the app. This is the single most important addition requested for police collaboration: it replaces anonymous, unverifiable alerts with a confirmed identity and a reachable emergency contact, so that a responding officer knows exactly who they are helping and who else to inform.

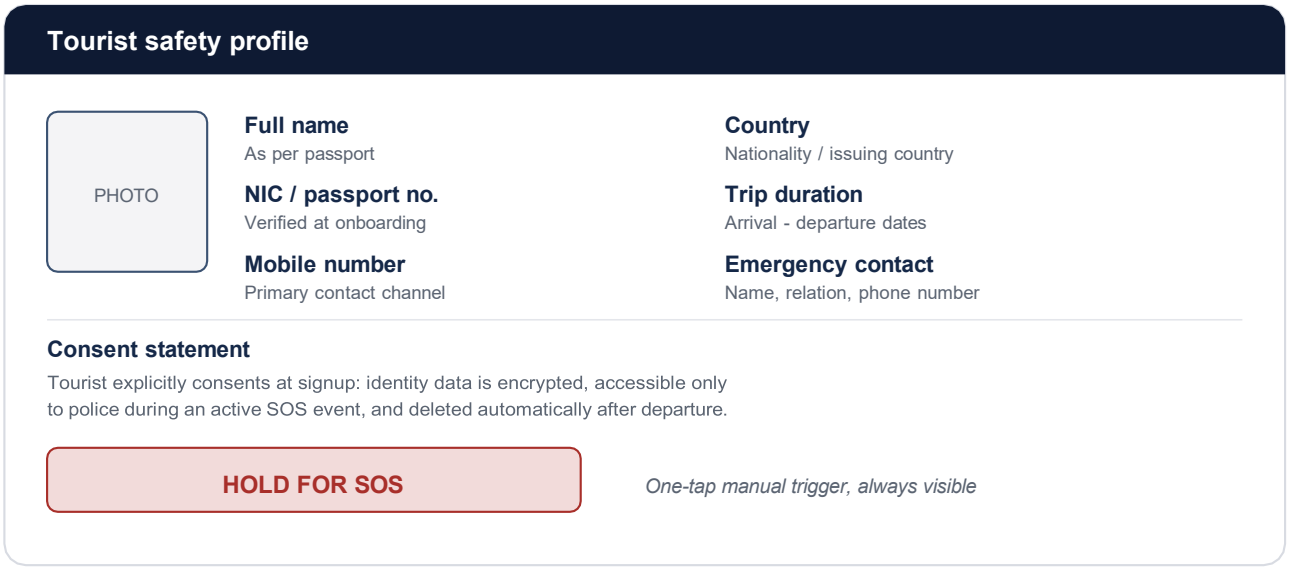


Figure 3: Registration profile captured at onboarding.

Field	Purpose for police response
Full name	Positive identification on arrival at the scene
Photo	Visual confirmation, especially useful in crowded or low-visibility locations
NIC / passport number	Official identity verification, cross-checkable with immigration records
Mobile number	Direct callback channel during and after dispatch
Country	Embassy or consulate notification if required
Trip duration (arrival-departure)	Confirms the tourist is currently in-country and active in the system
Emergency contact (name, relation, phone)	Immediate family notification, reducing distress and enabling follow-up

Data protection commitments

- Identity data is encrypted at rest and in transit; it is never publicly visible or shared with third parties.
- Full profile details (name, photo, NIC, emergency contact) are released to police systems only at the moment an SOS is confirmed - not before.
- Tourists give explicit, plain-language consent at registration, matching the ethical clearance approach already documented in the parent research proposal.
- Profiles are automatically purged after the registered trip end-date unless retained at the tourist's request.

Emergency Triggering and Police Response

Two independent trigger paths feed the same response pipeline, so an emergency is never missed because a tourist could not speak.

Trigger	How it works	Best suited for
Voice-activated SOS	Whisper speech-to-text and NER detect distress phrases across 10 languages, including Sinhala dialect	Situations where the tourist can speak, even briefly
Manual SOS button	A single, always-visible tap on the app's main screen, no confirmation dialogue required	Active threats, language barriers, or when speaking is unsafe

Once triggered, the system calculates the nearest active police station by GPS distance - not by a fixed jurisdiction boundary alone - and pushes the verified profile and live location to that station's dashboard within seconds.

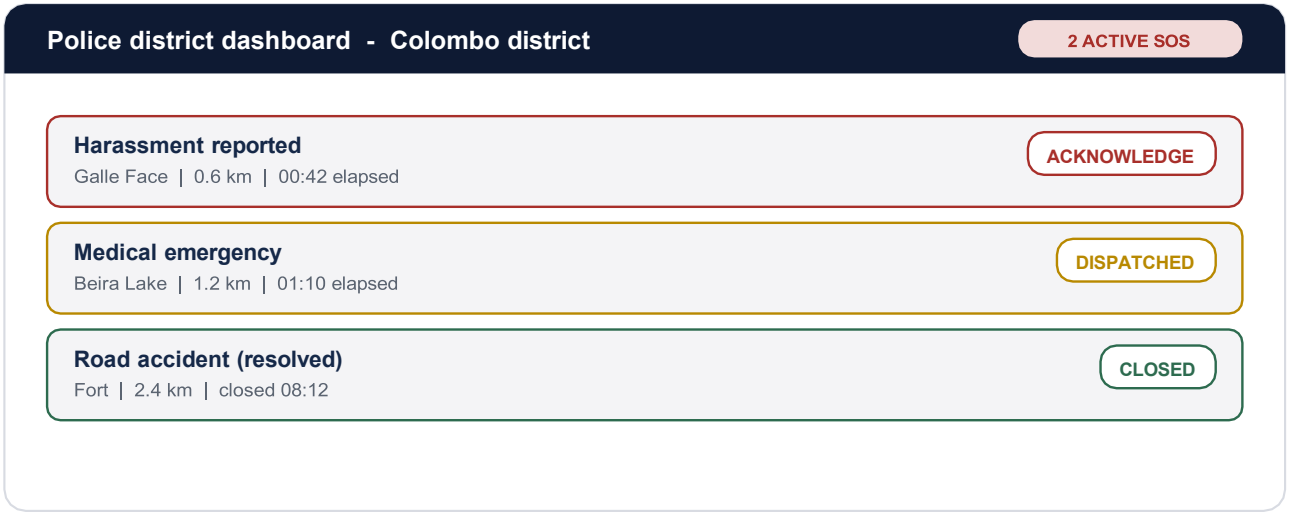


Figure 4: Police district dashboard - officers see active alerts, distance, and elapsed response time, and can acknowledge or escalate directly.

Dashboard capabilities for the Police Department

- District-based login, so each station only sees alerts relevant to its jurisdiction and surrounding radius.
- Live map view of active SOS events with incident type, tourist identity, and elapsed response time.
- One-tap acknowledge and escalate actions, feeding a response-time log usable for internal reporting.
- Parallel notification to the nearest verified hotel and to the tourist's registered emergency contact, so families are not left waiting.

Multilingual Support - Google Cloud Translation API

The distress detection pipeline transcribes tourist speech in the original language (Sinhala, Tamil, and eight other supported languages). For a police officer to act on that transcript immediately, it needs to reach the dashboard in English as well. The Google Cloud Translation API is proposed for this step, and is confirmed to be accessible for this project on the terms below.

Aspect	Detail
Availability	Yes - accessible through a standard Google Cloud Platform (GCP) account; no special approval needed for standard text translation
Free usage tier	500,000 characters per month, permanently free (does not expire, resets monthly)
Cost beyond free tier	USD 20 per million characters - at this project's expected volume (SOS transcripts and short alerts), monthly cost is expected to stay within or just above the free tier
Sinhala and Tamil support	Both are supported language codes, alongside 130+ other languages, with automatic language detection available
Setup requirement	A GCP project with billing enabled (a card is required even to use the free tier, but no charge occurs unless the free allowance is exceeded)
New account benefit	New GCP accounts receive a USD 300 trial credit valid for 90 days, useful for development and testing before pilot launch

Figures current as of mid-2026; Google Cloud pricing should be re-verified at implementation time via cloud.google.com/translate/pricing.

Where this fits in the pipeline

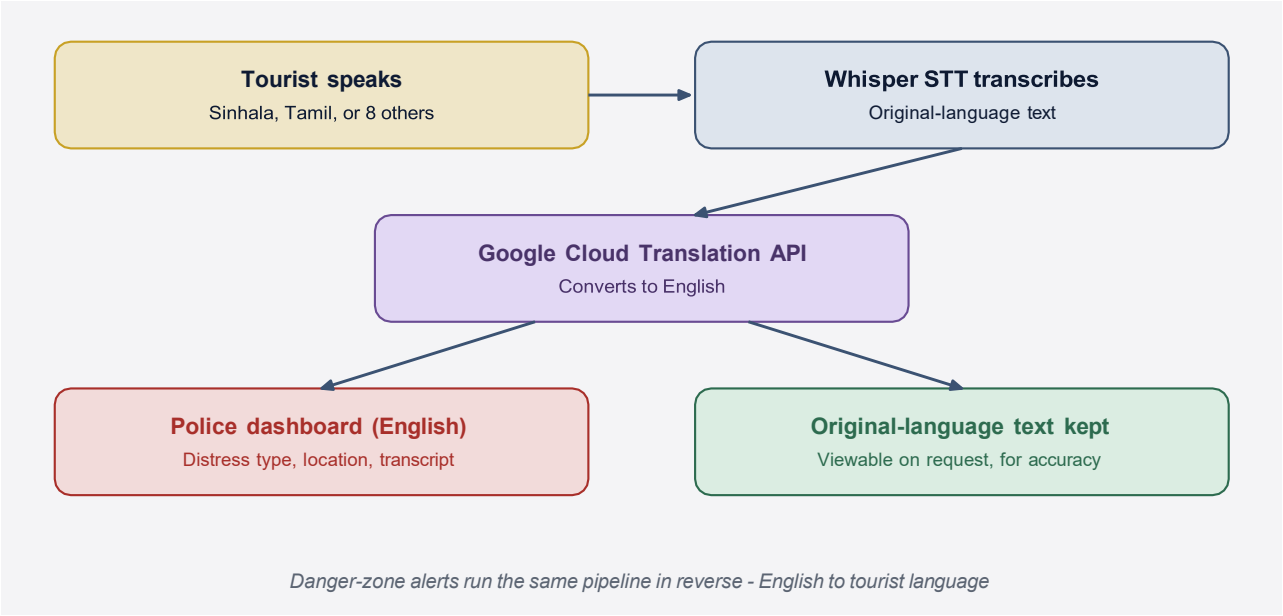


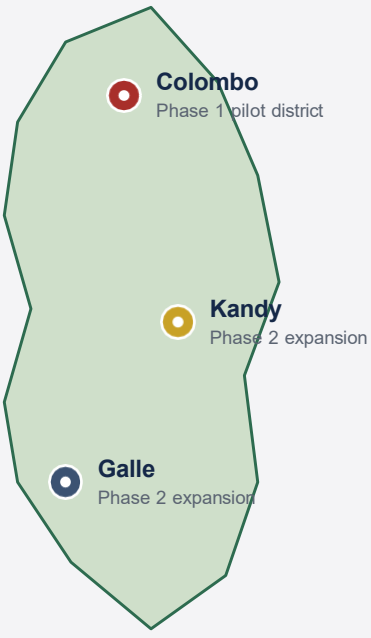
Figure 5: Translation step between voice transcription and the police dashboard, applied in both directions.

Why This Matters for the Police Department

Current challenge	How the system helps
Tourist emergencies reported late or not at all	Sub-90-second detection-to-dispatch target, with a manual fallback
Language barrier during distress calls	Automatic translation-ready transcription across 10 languages
Unverified or prank alerts	Every alert is tied to a registered, verified tourist profile
Difficulty locating the tourist	GPS-accurate location shared directly with the responding station
Family and embassy communication delays	Emergency contact and nationality notified automatically and in parallel

Proposed Next Steps

Phase	Activity	Outcome
1. Consultation	Joint review of this proposal with Police Department representatives	Feedback on dashboard workflow and data-sharing terms
2. Pilot design	Select 1-2 districts (e.g. Colombo, Galle) for a controlled pilot	Defined pilot scope and success metrics
3. Technical integration	Connect dashboard to existing station communication channels	Working alert-to-acknowledge pipeline
4. Field testing	Simulated SOS scenarios with officer feedback	Validated response times and usability
5. Rollout	Phased expansion to additional tourist districts	Nationwide tourist safety coverage



Illustrative pilot map (not to scale)

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